

Pathology Report

Confidential: For Authorized Personnel Only
Green Valley Diagnostic Laboratory
Report No.: HV45023947

Patient Information (CONFIDENTIAL)

Patient Name	Sarah L. Whitman	Home Address	23 Blueberry Drive, Greenfield, Oregon, USA
DOB	12 September 1985	Referring Physician	Dr. Raymond Carter
Medical Record No.	738493920	Physician Contact	+1 503-555-0182
Hospital No.	756 239 8192	Report Date	18 October 2023

Clinical History

The patient, Sarah Whitman, was referred due to symptoms of persistent abdominal discomfort, mild jaundice, and significant weight loss over three months. Imaging (MRI) indicated a mass in the hepatic region. Elevated liver enzymes (AST, ALT) and marginally raised AFP levels were noted. Family history of malignancy prompted this biopsy request. Previous cases include benign adenomas from 2018 (Ref: File No. C2738202).

Specimen Details

Specimen ID: HV45023947-A
Specimen Type: Liver biopsy
Procedure Date: 15 October 2023
Received Date: 16 October 2023

Macroscopic Description

Feature	Details
Dimensions	2.5 cm (length) x 0.3 cm (diameter)
Appearance	Firm, tan-brown tissue with slightly granular texture; no gross necrosis or hemorrhage
Observations	Nodular surface, tan color, and homogenous consistency
Special Notes	No encapsulation or defined margins, suggesting invasive nature

Microscopic Findings

Hematoxylin and Eosin (H&E) staining was performed, and slides reveal the following features:

- **Cellular Architecture:** The tissue displays a nodular pattern with dense cellularity, suggestive of a potential neoplastic process. Cells are predominantly polygonal with abundant eosinophilic cytoplasm.
- **Nuclear Features:** Nuclei appear pleomorphic, with occasional large nucleoli. Some cells display mitotic figures, although at a low rate.
- **Fibrosis and Vascularity:** A moderate degree of fibrosis is present, interspersed with small-caliber blood vessels. Areas with mild vascular proliferation and congestion are noted.
- **Infiltrative Edges:** The lesion shows poorly defined infiltrative margins, potentially extending beyond the biopsy core, indicative of invasive behavior.

Immunohistochemical Analysis

Immunohistochemistry was conducted using the following panel:

- **HepPar-1:** Positive, indicating hepatic origin.
- **CK7/CK20:** Negative, suggesting the lesion may be primary rather than metastatic from other adenocarcinomas.
- **Glypican-3:** Positive, which is often associated with hepatocellular carcinoma.
- **Ki-67 Proliferation Index:** Approximately 20%, indicating a moderately proliferative activity, consistent with a low-to-moderate grade malignancy.
- **p53 Mutation Analysis:** Mild aberrations were observed, but no definitive mutations indicating aggressive pathology were noted.

Additional Testing Recommendations

- **Genomic Testing:** Due to the family history of malignancy and presence of hepatic lesion, genomic sequencing could provide further insight into specific driver mutations.
 - **Repeat Imaging:** Follow-up MRI with contrast enhancement in 4-6 weeks to assess growth rate and lesion progression.
 - **Clinical Consult:** Immediate oncological referral for treatment options.
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Summary Diagnosis

Primary Diagnosis	Hepatocellular carcinoma (HCC), low-to-moderate grade with moderate fibrosis and low mitotic index
Clinical Stage	Likely early stage; infiltrative features suggest need for follow-up staging via imaging
Prognosis	Favorable with early intervention; close monitoring advised for potential invasive spread

Recommendations and Prognosis

Early intervention is suggested based on low-to-moderate grade findings. Options include surgical resection, depending on the lesion's proximity to vascular structures. Additional therapies such as localized ablation may be discussed following final staging.

Follow-Up:

- **Consultation with Oncologist:** Discuss further intervention options based on lesion staging.
 - **Genetic Counseling:** Due to family history and potential genetic predispositions.
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Authorization and Confidentiality Notice

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